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The Thickness of Mono-age Disk Populations Follows the Curvature of the Total Potential

Carlos Vera-Ciro[1]

[1] *Independent Researcher*

ABSTRACT

We test, population by population, whether the scale height of a mono-age disk population is set by its own vertical velocity dispersion in the total gravitational potential, as models of the Milky Way disk have long assumed, or by the self-gravitating sheet that studies of disk heating apply to single populations. In seven disks of the Auriga simulations, an isothermal tracer in a potential fitted to all the particles predicts the scale heights of 63 of 70 populations of 1 Gyr within 10 percent, near the floor that the vertical Jeans equation sets, whereas the sheet does so for 8. The heights grow as σ_z^{η} with η of 1.10 to 1.26, the local slope of the tracer of each potential, so age-thickness relations grow as age$^{\eta\beta_z}$ rather than the age$^{2\beta_z}$ of the sheet. A small residual for old populations, which persists with disk stars alone and in bins of 0.25 Gyr, an excess of mass far from the plane, and vertical actions with heavier tails than an isothermal population all reflect dispersions that rise with height. Summed over ages, the tracers reproduce the heights and the flare of each whole disk. In the solar neighborhood, Gaia DR3 star counts split by LAMOST and APOGEE ages follow the tracer with no trend with age in potentials that match the measured local densities, and imply midplane densities of 0.089 ± 0.017, 0.104 ± 0.013 and $0.120 \pm 0.030\, M_\odot\,\mathrm{pc}^{-3}$ with the three age scales. The published heights of Imig et al. (2025), fitted the same way, agree with our counts in level but grow more slowly with the dispersion than the tracer in any admissible potential.

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1. INTRODUCTION

The vertical structure of a population of stars of one age records how the disk heated it. Both its dispersion (J. Holmberg et al. 2009; Y.-S. Ting & H.-W. Rix 2019; S. Sharma et al. 2021) and its thickness grow with age, the thickness now measured population by population, split by abundance or by age, in large spectroscopic and astrometric surveys (J. Bovy et al. 2012b, 2016; J. T. Mackereth et al. 2017; D. Mirabal et al. 2026). Turning one into the other needs a model of the vertical structure, of which two are in use. In the first, a population of one age is an isothermal tracer of the total gravitational potential, whose density falls as $\exp[-\Delta\Phi(R,z)/\sigma_z^2]$ with the vertical potential $\Delta\Phi$ of all the mass. Models of the Milky Way disk have long been built this way, from a sum of such populations with dispersions prescribed by an age-velocity relation and compared with star counts that mix all ages (O. Bienaymé et al. 1987; A. C. Robin et al. 2003; A. Just & H. Jahreiß 2010; K. Sysoliatina & A. Just 2022). The density of a single tracer has been predicted from its kinematics in this way (J. N. Bahcall 1984; J. Holmberg & C. Flynn 2000). Mono-abundance populations, whose vertical dispersions change little with height (J. Bovy et al. 2012a), have been used in the inverse direction, to infer the potential from their densities and kinematics (J. Bovy & H.-W. Rix 2013).

In the second model, a population is a self-gravitating isothermal sheet of surface density Σ, with the profile $\mathrm{sech}^2(z/z_0)$ and $z_0 = \sigma_z^2/(\pi G\Sigma)$ (L. Spitzer 1942; P. C. van der Kruit & L. Searle 1981). The sheet describes the old disk as a whole, and studies of disk heating and radial migration apply it to single populations, mostly as a scaling between dispersion and thickness (R. Roškar et al. 2013; M. Martig et al. 2014; R. J. J. Grand et al. 2016; J. T. Mackereth et al. 2019). M. Martig et al. (2026, Sect. 5.3), for example, use it to turn a dispersion that grows as age$^{\beta_z}$ into an expected thickness that grows as age$^{2\beta_z}$. The two models agree for a population that makes its own potential, and differ for one that holds a small part of the mass of its disk. At fixed Σ the sheet makes every population grow as σ_z^2, whereas a tracer grows as σ_z^{η}, with η set by the shape of $\Delta\Phi$ over

the heights that the population reaches: $\eta = 1$ where the potential is harmonic and $\eta = 2$ where it grows linearly with height (P. C. van der Kruit 1988). Neither model has been tested population by population against heights measured independently of it, because the models of the Milky Way assume an age-velocity relation and fit counts that mix the populations, whereas the inverse problem leaves the heights free.

The question also bears on the relation between radial action and scale height that P. A. Palicio et al. (2025) find across the disks of the Auriga simulations. Its slope has been derived for a whole disk from the radial action of an epicycle and the scale height of a sheet (Y. Jia et al. 2023; C. Vera-Ciro, in press), so that the age dependence of the slope for a single population depends on which vertical model holds.

Here we test the tracer and the sheet population by population. In seven disks of the Auriga simulations (R. J. J. Grand et al. 2017, 2024), whose heights we measure from the star particles and whose potentials we fit to all the particles, we predict the scale height of every population of 1 Gyr of ages from its own dispersion, and compare the predictions with the measured heights, with the vertical Jeans equation as the floor of any such test. We measure how the heights grow with the dispersion, and compare that growth with the local slope of the tracer of each potential. In the solar neighborhood, we split star counts of Gaia giants into populations with the ages of LAMOST and APOGEE, and test the tracer in a family of potentials that match the measured density near the Sun. We describe the vertical structure of a tracer and our estimator in Section 2, the Auriga populations in Section 3, whole disks in Section 4, the Milky Way in Sections 5 and 6, and discuss the consequences in Section 7.

2. THE VERTICAL STRUCTURE OF A TRACER

2.1. *The tracer and the sheet*

A population whose distribution function depends on the vertical energy as $\exp(-E_z/\sigma_z^2)$ has, in the vertical potential $\Delta\Phi(z) = \Phi(R, z) - \Phi(R, 0)$ at a radius R, the density

$$\rho(z) \propto \exp\left[-\frac{\Delta\Phi(z)}{\sigma_z^2}\right], \tag{1}$$

with the same dispersion σ_z at every height. We call this population an isothermal tracer, and take $\Delta\Phi$ from the potential of all the mass. When the population makes the potential itself, the Poisson equation closes Eq. (1) into the sheet of L. Spitzer (1942),

$$\rho(z) \propto \operatorname{sech}^2\left(\frac{z}{z_0}\right), \qquad z_0 = \frac{\sigma_z^2}{\pi G \Sigma}. \tag{2}$$

Above z_0 the sheet falls as an exponential of scale $z_0/2$, which is why the relation is also written $\sigma_z^2 = 2\pi G\Sigma h$ for the scale h of that tail. We use z_0, which our estimator of the scale height returns for a sheet within 0.2 percent (Section 2.4), so that the convention does not enter the comparison. Applied to a population of one age, Σ is the surface density of all the mass near the plane, which we take within $|z| < 1.1$ kpc, so that the sheet predicts $h_Z \propto \sigma_z^2$ for every population of a disk.

2.2. *The local slope of the heights*

For a tracer in a potential $\Delta\Phi \propto |z|^p$, Eq. (1) gives a height that grows as $\sigma_z^{2/p}$. We define the local slope

$$\eta = \frac{d \ln h_Z}{d \ln \sigma_z}, \tag{3}$$

which is 1 in a harmonic potential, as near the midplane of any smooth distribution of mass, and 2 in a potential that grows linearly with height, as above a thin one (P. C. van der Kruit 1988). A real disk has a harmonic core and a linear rise, so η grows with σ_z from 1 toward 2 as the populations reach heights where the potential is no longer harmonic. For a dispersion that grows with age as age^{β_z}, a tracer thickens as $\text{age}^{\eta\beta_z}$ over a range of σ_z in which η is constant, whereas the sheet at fixed Σ gives $\text{age}^{2\beta_z}$. Far above the plane, where a dark halo dominates, $\Delta\Phi$ grows more slowly than $|z|$, so η would exceed 2 there, although Section 3.3 shows that the profiles lose their exponential scale first.

2.3. *What a test of the tracer can show*

Part of any agreement between the tracer and the measured heights is guaranteed. In a harmonic potential $\Delta\Phi = \nu^2 z^2/2$, the vertical virial theorem of a steady population gives $\langle v_z^2 \rangle = \nu^2 \langle z^2 \rangle$ whatever its distribution function, so the spread of a cold population in $|z|$ follows from that of its vertical velocities, as Eq. (1) gives it. A cold population whose dispersion near the plane represents its stars is therefore bound to agree with the tracer, to the extent that a fitted scale height measures the spread in $|z|$. The test bites for the hot populations, which reach heights where the potential is no longer harmonic and where the shape of the distribution function sets the profile. We therefore report the cold and the hot populations separately.

The tracer also assumes that σ_z does not change with height and that the population feels no radial gradient of its motions, the tilt term of the vertical Jeans equation. We integrate that equation with the dispersion and the tilt term measured at every height, which holds for any steady axisymmetric population, so that its scatter about the measured heights sets the floor of any prediction from the kinematics at one radius. The same two

refinements improve estimates of the local density from one tracer in N-body disks (S. Garbari et al. 2011; S. Sivertsson et al. 2022).

2.4. *The scale height and its window*

We measure the scale height of a set of heights $|z|$, or of a profile, by maximum likelihood of an exponential truncated at a window L. Its mean height inside the window is

$$\langle z \rangle = h - \frac{L}{e^{L/h} - 1}, \tag{4}$$

which we solve for h. The maximum-likelihood estimate of the rate $1/h$ exists only when $\langle z \rangle < L/2$ (W. L. Deemer & D. F. Votaw 1955, Sect. 3), so a window whose mean reaches $L/2$ gives no height. As P. A. Palicio et al. (2025) do, we average the estimates within L and within $L/2$. A profile that is not exponential has no single scale, so its fitted height depends on the window: a window narrower than the profile reads its flat core as a large scale (Appendix A). We therefore scale the window with the population, $L = 2h_Z$, taking the smallest L that satisfies this condition, so that populations of one profile shape are fitted over the same part of their profiles. An exponential returns its scale in any such window, whereas the sheet of Eq. (2) returns z_0 within 0.2 percent. A window needs 100 stars inside $L/2$: with fewer, the profile of a hot population is nearly flat inside a small window, so that the noise of the fitted $1/h$ can satisfy the condition by chance far below the true scale (Appendix A). Every prediction passes through the same estimator as the stars. Because no window is free of convention, we also measure the half-mass height, below which half of the mass within $|z| < 12\,\mathrm{kpc}$ lies.

3. MONO-AGE POPULATIONS IN AURIGA

3.1. *Simulations and measurements*

We use the seven halos of the Auriga simulations (R. J. J. Grand et al. 2017) at the standard resolution of the public release (R. J. J. Grand et al. 2024) that P. A. Palicio et al. (2025, Table 1) analyze, Au2, Au3, Au16, Au19, Au23, Au24 and Au27, at redshift zero. For each halo we fit an axisymmetric potential to all the particles of the snapshot with AGAMA (E. Vasiliev 2019), as a CylSpline expansion for the gas and the stars of the disk region and a Multipole expansion for everything else. The star particles of the annulus $7.5 < R < 9\,\mathrm{kpc}$ are split into ten bins of 1 Gyr in age, from 0 to 10 Gyr, which give 70 populations. In each bin, σ_z is the dispersion of v_z of the stars with $|z| < 0.3\,\mathrm{kpc}$, clipped at four standard deviations estimated from the median absolute deviation, whereas h_Z is the scale height of the heights of all its stars, weighted by their initial mass, in the window of Section 2.4. The tracer of Eq. (1) takes $\Delta\Phi$ from the fitted potential at the median radius of the annulus, whereas the sheet of Eq. (2) takes Σ from all the particles of the annulus within $|z| < 1.1\,\mathrm{kpc}$. Errors are half the range between the 16th and the 84th percentiles of 300 bootstrap resamplings of the stars of each bin, each of which recomputes every quantity, so that the ratio of a predicted to the measured height carries the correlation of the two. On a mock snapshot of a known galaxy, the analysis recovers $\Delta\Phi$ within 1.0 percent and the height of every bin within 0.2 percent (Appendix B). In the real snapshots, the potential averaged over azimuth gives tracer heights 0.97 to 1.04 times those of the fit.

3.2. *The tracer, population by population*

The tracer predicts the height of 63 of the 70 populations within 10 percent, 44 within 5 percent, with ratios of 0.83 to 1.16 and a median of 1.01 (Figure 1). The populations thus have the heights that their own dispersions give them in the potential of the whole galaxy, from the youngest to the oldest, whose heights reach 4.6 kpc with σ_z of up to $127\,\mathrm{km\,s^{-1}}$.

As Section 2 argued, part of this agreement holds for any steady population. Below $h_Z = 1\,\mathrm{kpc}$, where the potential is close to harmonic, the tracer comes within 10 percent in 30 of 32 populations, with a scatter of 5.1 percent. Above 1.5 kpc it does in 22 of 26, with a scatter that grows to 9.5 percent above 2.5 kpc. Of the populations hotter than $60\,\mathrm{km\,s^{-1}}$, which are closer to a spheroid than to a disk, the tracer predicts 16 of 19 within 10 percent, against 47 of 51 of the disk populations, with a scatter of 5.5 percent. Section 3.6 repeats the test with the stars on disk orbits alone.

3.3. *The local slope of the heights*

We fit η of Eq. (3) as the slope of the weighted least-squares line of $\ln h_Z$ against $\ln \sigma_z$ across the bins of each halo. Both coordinates carry errors, so each bin is weighted by the inverse of $(\delta h_Z/h_Z)^2 + \eta^2(\delta\sigma_z/\sigma_z)^2$, refitted until η converges. The error is scaled by $\sqrt{\chi^2_\nu}$ where $\chi^2_\nu > 1$. Over 0-10 Gyr, η spans 1.10 to 1.26 across the seven halos, with errors of 0.03 to 0.05, between the value of 1 of a harmonic potential and the value of 2 of a linear one. Because η is a local slope, it changes with the range of ages, to 1.03 to 1.35 over the ranges 0-10, 2-10, 0-8 and 2-8 Gyr.

Figure 2 compares η with the tracer itself. For each halo we compute the tracer of the fitted potential at 41 dispersions from 5 to $150\,\mathrm{km\,s^{-1}}$, measure its height in the window, and take the logarithmic derivative. The local slope rises from 1.01 to 1.03 at $10\,\mathrm{km\,s^{-1}}$, where

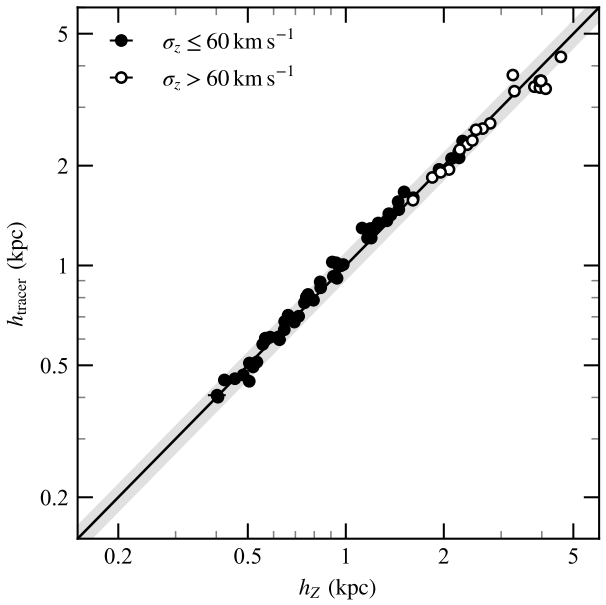


Figure 1. The scale height that the isothermal tracer predicts for every population of the seven Auriga halos, against the measured h_Z with its error. Filled circles are the populations with $\sigma_z \leq 60\,\mathrm{km\,s^{-1}}$, hollow circles the hotter ones. The line marks equality, inside a gray band of agreement within 10 percent.

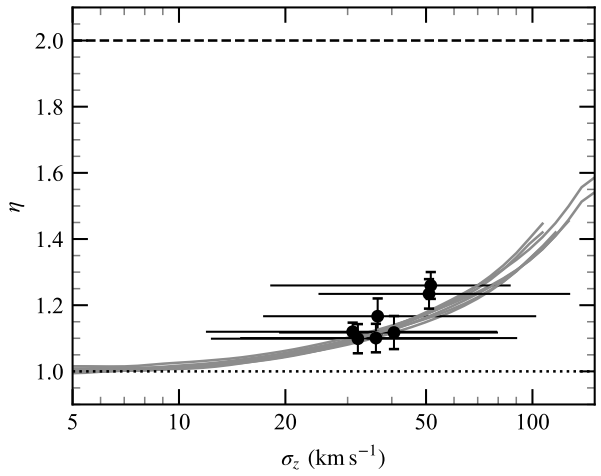


Figure 2. The local slope $\eta = d\ln h_Z / d\ln \sigma_z$ of the tracer of each halo's fitted potential (gray lines), and the η fitted to the populations of 0-10 Gyr of each halo (circles with errors), drawn across the σ_z of those populations. The dotted and dashed lines mark the η of a harmonic and of a linear potential.

the populations stay inside the harmonic core of the potential, to 1.11 to 1.13 at $40\,\mathrm{km\,s^{-1}}$, and to at most 1.38 to 1.59 at the hottest dispersions. Over the dispersions of the populations of each halo, the slope of the chord of this curve differs from the measured η by -0.03 to 0.11, within two standard errors in 6 of 7 halos. The exception is Au19, whose η of 1.26 ± 0.04 exceeds the chord of 1.15, because its old populations are thicker than the tracer allows (Section 3.4). The slope stays below 2 over the whole grid: in 5 of 7 halos the window finds no scale above 127 to $150\,\mathrm{km\,s^{-1}}$, where the tracer is so flat inside $2h_Z$ that its mean height reaches $L/2$. The hottest population of each of these halos lies a factor of at least 1.4 below that limit. The heights of the populations therefore grow with their dispersions as the curvature of each potential prescribes.

3.4. *Age dependence and the Jeans equation*

The age dependence of the heights tests the tracer more strictly than its scatter does, because a prediction can scatter about the measured heights without a trend, or follow them in scatter but miss how they change with age. We fit the trend p of $h_{\mathrm{pred}}/h_Z \propto \mathrm{age}^p$ by weighted least squares over the bins of each halo, with the error scaled by $\sqrt{\chi^2_\nu}$. A prediction that reproduces how h_Z changes with age has $p = 0$. Over 0-10 Gyr the trend of the tracer lies within 1.6 standard errors of 0 in every halo, with values of -0.045 to 0.014 (Figure 3). Over 2-10 Gyr it is negative in 7 of the 7 halos, a split of signs with a probability of 0.016 under a scatter symmetric about 0. The tracer therefore falls slightly short of the thickness of the oldest populations, by an amount that only some halos resolve on their own.

The vertical Jeans equation locates that residual. With the variance of v_z of each population measured in slices of $|z|$ up to 12 kpc, without clipping, it gives heights within 10 percent in 61 populations. Its trends over 2-10 Gyr are negative in 4 of the 7 halos, so the pattern of signs of the tracer is gone. The exception is Au19, whose populations older than 5 Gyr reach 0.84 to 0.91 of their measured heights in these Jeans heights. Their tilt term, the radial derivative of $\nu\sigma_{Rz}$ that we fit over $6 < R < 10.5$ kpc, opposes the vertical force between 1 and 5 kpc, and brings them to 0.96 to 1.08. Their mean vertical motions, with thickening rates of -0.30 to 0.08 per Gyr, show no more departure from a steady state than those of the other halos. With the tilt term, the Jeans heights lie within 10 percent in 66 populations, with ratios of 0.82 to 1.16 and a scatter of 5.4 percent. Because the Jeans equation with the tilt term holds for any steady axisymmetric population, that scatter measures what a test at one radius cannot remove, departures from a steady state and the noise of the measured profiles, and so sets the floor of any prediction of these heights. The tracer's scatter of 6.2 percent lies just above it.

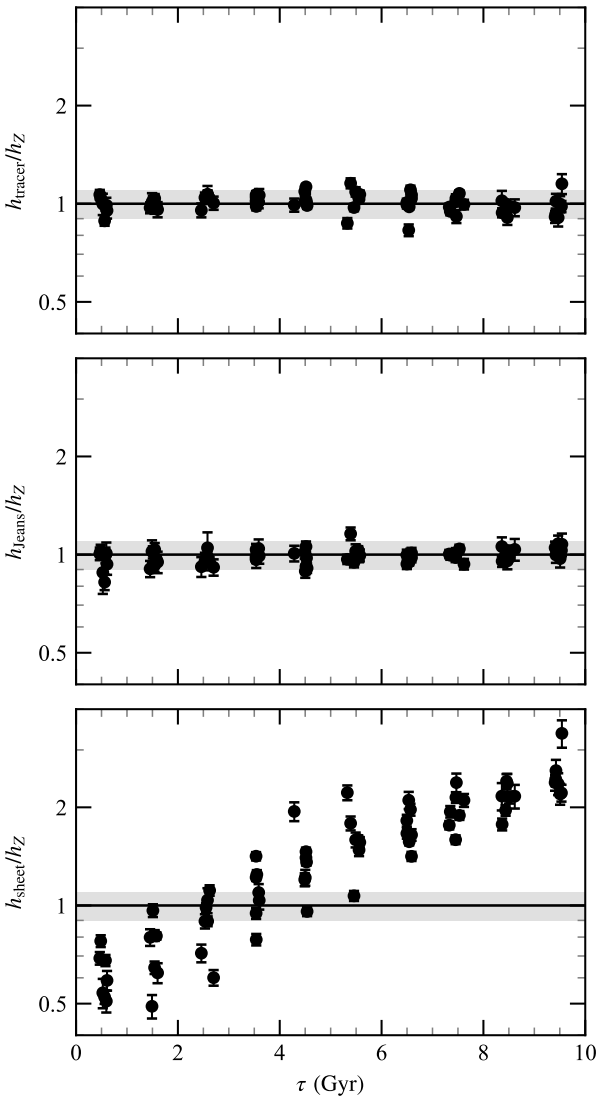


Figure 3. Predicted over measured h_Z for every population against its median age: the isothermal tracer (top), the Jeans equation with the measured $\sigma_z(z)$ and the tilt term (middle), and the sheet (bottom), with their bootstrap errors. The gray band marks agreement within 10 percent.

3.5. *Half-mass heights*

The half-mass height needs no window, and weighs the heights above $2h_Z$ that the window leaves out. The tracer's half-mass heights lie within 10 percent of the measured ones in 60 of 70 populations, with a median ratio of 0.94 and a scatter of 7.8 percent, and exceed them in only 1 of 70. The shortfall is nearly the same at every age: the trends of the ratio over 0-10 Gyr lie within 2.9 standard errors of 0. With half-mass heights in place of h_Z, η is 1.15 to 1.28. The populations of every age therefore carry more of their mass far from the plane than the isothermal profile of their midplane dispersion. The Jeans heights trace the excess to the rise of σ_z with height: with the measured $\sigma_z(z)$ their half-mass heights lie within 10 percent in 64 of 70 populations, and with the tilt term in 69 of 70, with a scatter of 3.6 percent. The sheet comes within 10 percent in 8 of 70, with a median of 1.32, so its failure does not come from the window.

3.6. *Disk stars, finer bins and clipping*

Three checks ask whether the residual over 2-10 Gyr comes from how the populations are selected and measured (Figure 4). Each repeats the measurement in its own bins with 100 bootstrap resamplings. Keeping only the stars whose circularity $L_z/L_c(E)$ exceeds 0.7 keeps 0.07 to 1.00 of the mass of a bin, the least in the oldest bins. With these stars alone, the tracer predicts h_Z within 10 percent in 62 of 70 bins. Its trends over 0-10 Gyr lie within 0.9 standard errors of 0, whereas over 2-10 Gyr 7 of 7 are negative. In bins of 0.25 Gyr, 6 of 7 of the trends over 2-10 Gyr are negative, up to 4.8 standard errors from 0. Without clipping, σ_z keeps the tails of v_z, which raise the tracer heights by 0 to 24 percent. The tracer of the unclipped dispersion comes within 10 percent in 52 of 70 bins, with trends over 2-10 Gyr that are negative in 7 of 7 halos. The residual is therefore a property of the disk stars, persists in populations of 0.25 Gyr, and does not come from the clipping, which leaves the rise of σ_z with height as its cause.

3.7. *The shape of the vertical actions*

Y.-S. Ting & H.-W. Rix (2026) describe a coeval population by the distribution of its vertical actions, $p(J_z) \propto \exp[-(J_z/J_0)^{2-b}]$, which diffusion with a coefficient $\propto J_z^b$ leaves behind. Scatterers that fill the volume, such as bending waves or dark substructure, give $b = 1$, whereas scatterers confined to a thin layer at the midplane, such as giant molecular clouds, give $b = 1/2$, close to the $b = 0.51$ they measure for red-clump stars of 2-8 Gyr in the Milky Way. M. Djurić & R. Schönrich (2026) likewise derive vertical distribution functions in the actions of coeval populations scattered by giant molecular clouds, which depart from the isothermal shape. We fit b and J_0 to the vertical actions of the stars of each bin by maximum likelihood, weighted by initial mass, with the actions from the Stäckel fudge of AGAMA in the fitted potential (J. Binney 2012). An isothermal population is exponential in J_z only where the potential is harmonic: in a potential $\propto |z|^p$ its energy grows as $J_z^{2p/(p+2)}$, so a linear potential gives it $b = 4/3$. We therefore read each fitted b against $b_{\rm iso}$, the b of the isothermal tracer of the bin's σ_z in the same

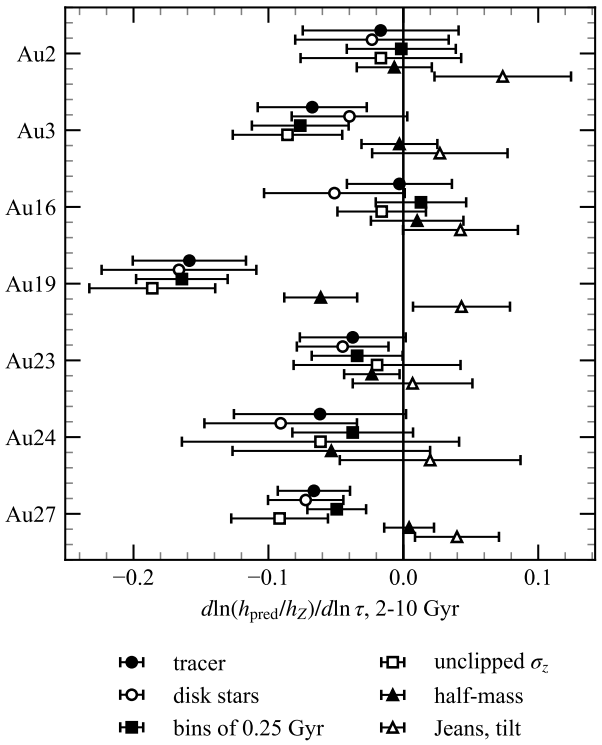


Figure 4. The trend $d\ln(h_{\rm pred}/h_Z)/d\ln\tau$ of the tracer over 2-10 Gyr in each halo: in the main measurement (filled circles), with the stars on disk orbits alone (hollow circles), in bins of 0.25 Gyr (filled squares), with the unclipped σ_z (hollow squares) and for half-mass heights (filled triangles), and the trend of the Jeans heights with the tilt term (hollow triangles).

potential, fitted to its distribution of actions in the same way. Both need the orbits of the population to stay inside the 12 kpc of the grid of heights, which those of 52 of 70 populations do.

The fitted b is 1.02 to 1.60 and exceeds $b_{\rm iso}$ in 52 of 52 populations, by a median of 0.36. The cold populations have b of 1.23 to 1.57, against $b_{\rm iso}$ of 1.02 to 1.11. The vertical actions of the Auriga populations thus have heavier tails than an isothermal population of the same midplane dispersion, as their half-mass heights showed, and depart from it in the direction opposite to the thin layer. The population with $b = 1/2$ makes the same point through the heights. With J_0 set so that its stars within 0.3 kpc of the plane have the σ_z of the bin, its flat-topped velocities make it thicker than the tracer near the plane. It predicts h_Z within 10 percent in 24 of 52 populations, with a median ratio of 1.11, whereas over the same populations the tracer does so in 48 of 52, with a median of 1.02. The Auriga populations are therefore not thin-layer populations, which is not surprising in hindsight, because the star-forming gas of these simulations follows an effective equation of state rather than forming giant molecular clouds (R. J. J. Grand et al. 2017).

3.8. *The sheet*

Against the measured heights, the sheet comes within 10 percent in 8 populations, with ratios of 0.49 to 3.37: 0.51 to 0.78 for the youngest population of each halo and 2.20 to 3.37 for the oldest (Figure 3). Its trends with age over 0-10 Gyr are 0.40 to 0.82, at least 5 standard errors from 0 in every halo, as expected of a ratio that grows as $\sigma_z^{2-\eta}$, or age$^{(2-\eta)\beta_z}$ at fixed Σ. The sheet thus fits a population only where $\sigma_z^2/(\pi G\Sigma)$ happens to cross its height, at intermediate ages, making the young populations too thin and the old ones too thick. This is not surprising in hindsight: the sheet is the tracer of a layer whose own mass makes its potential, whereas a population of one age is one part of that layer and moves in the potential of all of it.

4. WHOLE DISKS

The relation of P. A. Palicio et al. (2025) uses the height of all the stars of a disk, which mixes populations of every age. If every population is an isothermal tracer, the profile of the whole disk is their sum, whose height follows from the kinematics of its populations and the potential alone. Models of the Milky Way disk are built this way, from isothermal components of different ages with dispersions prescribed by an age-velocity relation (A. C. Robin et al. 2003; A. Just & H. Jahreiß 2010; K. Sysoliatina & A. Just 2022). Here the dispersions are measured rather than prescribed, so that the sum can be tested against heights measured independently. We split the radial range of P. A. Palicio et al. (2025, Sect. 3.4) of each halo into rings of 1 kpc, and each ring into bins of 1 Gyr from 0 to 14 Gyr. Each cell enters as the tracer of its clipped σ_z within 0.3 kpc of the plane, in the fitted potential at the center of its ring, with the current mass of its stars, which is how P. A. Palicio et al. (2025) weight their histograms. A cell with fewer than 30 stars within 0.3 kpc of the plane has no σ_z and is left out, which removes at most 0.8 percent of the mass of a disk. Both the sum and the stars are fitted in the window of Section 2.4, with errors from 50 bootstrap draws of the stars.

One tracer per ring, with the dispersion of all its stars, predicts 0.97 to 1.13 of the measured height of the whole disk, the excess expected of one isothermal component for a mixture of dispersions. The sum over rings and ages predicts 0.92 to 1.04, with a median of 0.979. Ring by ring, it comes within 10 percent of the measured

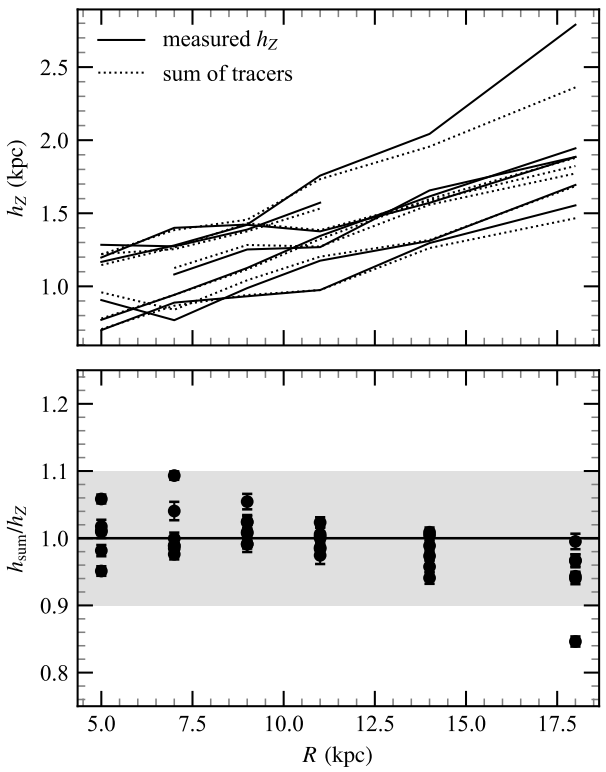


Figure 5. Top: the measured h_Z of the whole disk in every ring of 1 kpc of the seven halos (solid) and that of the sum of the tracers of its populations (dotted). Bottom: their ratio, with a gray band of agreement within 10 percent.

height in 38 of the 39 radial ranges, following the flare of every disk (Figure 5). The heights of whole disks, including those that enter the relation across galaxies, thus follow from the tracers of their populations, in the window in which the tracer was tested.

5. THE MILKY WAY

5.1. *Star counts split by age*

A spectroscopic survey cannot give the density of a population against $|z|$ without its selection function, whereas a complete photometric sample can. We count the Gaia DR3 (Gaia Collaboration et al. 2016, 2023) giants of a cylinder of radius 1.5 kpc around the Sun, within $|z| < 1.2$ kpc, whose reddening-free Wesenheit absolute magnitude $M_W = G - 1.9\,(G_{\rm BP} - G_{\rm RP}) + 5\log_{10}(\varpi/{\rm mas}) - 10$ lies in a window, and split each slice of $|z|$ into populations with the fraction of its spectroscopic stars that belong to each. The primary red-clump stars of LAMOST (Y. Huang et al. 2020) split the 461,601 Gaia stars within 0.3 mag of the red-clump peak of M_W, through 22,771 matches by position and G. APOGEE DR17 (S. R. Majewski et al. 2017; Abdurro'uf et al. 2022) splits the 745,901 giants of the box $-2.6 < M_W < -0.9$ through matches by Gaia source identifier, with the ages of astroNN (J. T. Mackereth et al. 2019) (16,057 matches) and those of DistMass (A. Stone-Martinez et al. 2024), which J. Imig et al. (2025) adopt (15,933 matches). APOGEE selects its targets by magnitude and color rather than by age (G. Zasowski et al. 2013, 2017). As long as the targeting of a survey does not depend on age at a fixed position, the fraction of a population in a slice is unbiased even though the completeness of the survey changes with $|z|$, an assumption that Section 5.6 tests for both surveys. A population is a range of 2 Gyr of ages of the low-[α/Fe] disk with [Fe/H] > -0.6. Its fraction varies across the cylinder as e^{gx}, with x the distance toward the Galactic center and one gradient g per population fitted to its matches.

We fit h_Z to the counts in the window of Section 2.4. The kinematics come from the giants of each population in the solar annulus $7.5 < R < 9$ kpc: σ_z from those with $|z| < 0.3$ kpc, clipped and with their median velocity error removed in quadrature, with every star in the Galactocentric frame of astroNN, the Sun at R_0 =8.125 kpc and 20.8 pc above the plane (M. Bennett & J. Bovy 2019). The tracer of each population, in the potential of P. J. McMillan (2017) at R_0, goes through the same counts, slice by slice, as do the sheet and the Jeans equation, so that a predicted and a measured height are fitted the same way. Errors come from 300 bootstrap draws of the Gaia stars, their matches and the kinematic stars.

5.2. *The tracer and the Jeans equation*

The tracer predicts the counts with ratios of 0.85 to 0.99 with LAMOST, 0.89 to 1.14 with APOGEE on the astroNN scale and 0.92 to 1.06 on the DistMass scale, with trends over 0-10 Gyr of $+0.019 \pm 0.082$ ($+0.2$ standard errors), $+0.016 \pm 0.044$ ($+0.4$ standard errors) and -0.002 ± 0.062 (-0.0 standard errors), and η of 1.10 ± 0.17, 1.09 ± 0.11 and 1.17 ± 0.16, within the range of the Auriga disks (Figure 6). It falls below the counts, however, in 13 of 15 bins, with medians of 0.92 to 0.94 against 1.01 in Auriga. Two causes could make it fall short: populations that are not isothermal, and a potential that is too dense near the plane. The Jeans equation with σ_z^2 measured in the slices 0-0.3, 0.3-0.6, 0.6-1.0 and 1.0-1.5 kpc, without the tilt term, which this sample cannot measure, gives heights of 0.81 to 1.12 of the counts, with medians of 0.88 to 0.91. The change of σ_z with height thus leaves the shortfall in place.

5.3. *A family of potentials*

The potential accounts for the shortfall. At R_0, the potential of P. J. McMillan (2017) has a mid-

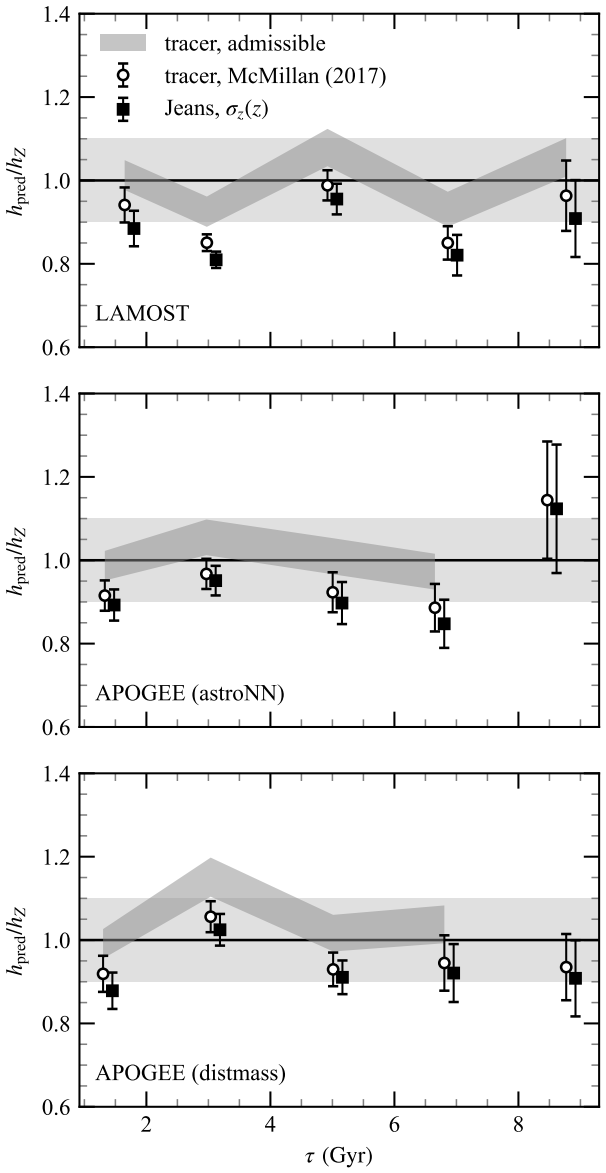


Figure 6. The tracer over the star counts of each survey against median age, in the potential of P. J. McMillan (2017) (hollow circles) and across the admissible potentials of Section 5.3 (gray band), and the Jeans equation with the measured $\sigma_z(z)$ (filled squares).

plane density of $0.117\,M_\odot\,\mathrm{pc}^{-3}$ and $77.4\,M_\odot\,\mathrm{pc}^{-2}$ within $|z| < 1.1\,\mathrm{kpc}$, above the $\rho_0 = 0.097 \pm 0.013\,M_\odot\,\mathrm{pc}^{-3}$ of C. F. McKee et al. (2015, their Eq. 57) and the $\Sigma_{1.1} = 68 \pm 4\,M_\odot\,\mathrm{pc}^{-2}$ of J. Bovy & H.-W. Rix (2013). Where the potential is harmonic the height of a tracer scales as $\rho_0^{-1/2}$, so a midplane density 20 percent above 0.097 thins a cold population by about 9 percent. We vary the surface density of the thin disk of P. J. McMillan (2017) by factors of 0.7, 0.8, 0.9 and 1.0, its scale height over 0.25, 0.30 and 0.35 kpc, and the normalization of its dark halo by factors of 0.7, 1.0 and 1.3. Of these potentials, 8 of 36 lie within one error of both measurements, with ρ_0 of 0.094 to 0.108 and $\Sigma_{1.1}$ of 66.8 to 70.7. In them the medians of the tracer over the counts are 0.97 to 1.07. The shortfall of the tracer therefore comes from the densities of the potential near the Sun rather than from the populations. The family is not refitted to the rotation curve, so its circular velocities at R_0, 213 to 224 km s^{-1}, lie below the 233 km s^{-1} of P. J. McMillan (2017). The tracer needs only the vertical potential near the plane, which ρ_0 and $\Sigma_{1.1}$ bound.

5.4. *The local potential from the tracer*

The family also turns the test around. Every bin carries the ratio of the tracer to its h_Z in each of the 36 potentials. Across the family, the logarithm of that ratio is a linear function of $\ln\rho_0$ and $\ln\Sigma_{1.1}$ to within 1.1 percent in every bin. If the populations are isothermal tracers, one potential brings every bin of a survey to its measured h_Z, which a weighted least-squares fit of the bins finds. This is the inverse use of the tracer that J. Bovy & H.-W. Rix (2013) made with mono-abundance populations, with the family standing in for a parameterized potential. The midplane densities are 0.089 ± 0.017, 0.104 ± 0.013 and $0.120 \pm 0.030\,M_\odot\,\mathrm{pc}^{-3}$ with LAMOST and with APOGEE on the astroNN and the DistMass scale (Figure 7), against the $0.097 \pm 0.013\,M_\odot\,\mathrm{pc}^{-3}$ that C. F. McKee et al. (2015) find by combining their census of stars, remnants and gas with a density of dark matter inferred from dynamical measurements of the surface density. $\Sigma_{1.1}$ is not constrained, because the populations stay below about 1 kpc, where ρ_0 sets the potential. The χ^2 per degree of freedom is 1.0 to 4.8, so with LAMOST and on the DistMass scale the bins scatter about one potential by more than their errors, which the quoted errors include.

5.5. *The shape of the Milky Way populations*

In the potential of P. J. McMillan (2017), at the dispersions of our bins, the isothermal tracer has $b_{\rm iso}$ of 1.10 to 1.17, so the $b = 0.51$ of Y.-S. Ting & H.-W. Rix (2026) departs from it by about 0.6. The population with $b = 1/2$ and the σ_z of each bin, through the same counts, gives medians of 1.03 to 1.05 in the potential of P. J. McMillan (2017) and 1.08 to 1.20 in the admissible potentials, against 0.92 to 0.94 and 0.97 to 1.07 for the isothermal tracer. Inverted as above, it needs midplane densities of 0.117 ± 0.034, 0.134 ± 0.018 and $0.146 \pm 0.071\,M_\odot\,\mathrm{pc}^{-3}$. Being thicker than the isothermal population of the same midplane dispersion, it fits the counts only in a potential denser than the measurements near the Sun, close to that of P. J. McMillan

(2017). At the measured densities the counts favor the isothermal shape, although the error of ρ_0 keeps the preference weak, so the star counts of the solar neighborhood do not by themselves decide the shape that Y.-S. Ting & H.-W. Rix (2026) measure in the actions.

5.6. *Age errors and selection*

The ages carry typical errors of 30 percent for the LAMOST red clump (Y. Huang et al. 2020) and up to 40 percent by our cut for APOGEE, so bins of 2 Gyr mix their neighbors. The targeting of either survey could also depend on age at a fixed position. Two checks remeasure every survey, with 100 bootstrap draws per bin. The first draws every age again from its error, lognormal with the relative error as the dispersion of ln age, which doubles the variance of the errors, in 4 realizations. The medians of the tracer over the counts become 0.89 to 0.95, against 0.92 to 0.94 with the ages as given. Its trends over 0-10 Gyr become -0.006 to 0.089, within 1.8 standard errors of 0. Doubling the errors of the ages thus leaves the test where it was, presumably because the mixing of ages enters the counts and the kinematics of a bin alike.

The second asks whether the fraction of a population at a fixed position depends on the magnitude of its stars, as it would if the targeting depended on age there. In each slice of $|z|$ we split the matches of each survey at their median G, and measure each survey with each half. The heights of the bright over the faint half are 0.81 to 1.29, within two standard errors in 13 of 15 bins and at most 2.7 standard errors apart: 5 of 5 with LAMOST, and 8 of 10 with APOGEE on either age scale. A targeting of APOGEE that depends on age at a fixed position may thus move some of its bins by up to the largest of these differences, whereas the LAMOST counts show no such dependence.

6. THE HEIGHTS OF IMIG ET AL.

J. Imig et al. (2025) fit every population of APOGEE DR17 with more than 100 stars, in bins of [α/M], [M/H] and DistMass age, with a model of the whole footprint of the survey corrected for its selection function. Their model of a population is $\nu(R,z) = \nu_\odot \Sigma(R) \exp[-|z|/h_Z(R)]$, with a broken exponential $\Sigma(R)$ and a scale height that flares linearly. Their heights at the Sun are therefore the scales of exponentials fitted to the whole range of $|z|$ that APOGEE reaches, whereas ours are windowed heights of counts below $|z| = 1.2$ kpc. An isothermal population is flat at the plane and falls more steeply above, so the exponential that matches its mean height is thinner than its windowed height: for the tracer of P. J. McMillan (2017) at R_0, the windowed height is 1.21 to 1.51 times the exponential fit below 1.2 kpc. We therefore compare each of their populations with our counts of the same cell of age and [M/H], both fitted with the same exponential below $|z| = 1.2$ kpc, their model put through our cylinder and counts. Of their cells, 40 of 101 have at least 100 matched stars.

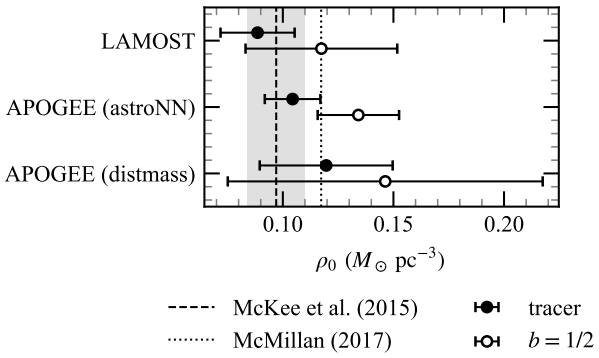


Figure 7. The midplane density ρ_0 at which every bin of a survey matches its counts, with one-sigma errors, for the isothermal tracer (filled) and for the population with $b = 1/2$ (hollow). The gray band and the dashed line mark the $0.097 \pm 0.013\, M_\odot$ pc^{-3} of C. F. McKee et al. (2015). The dotted line marks the ρ_0 of P. J. McMillan (2017).

Fitted the same way, our counts over their model have a median of 1.05 over 40 cells, and change by $+0.033 \pm 0.085$ (+0.4 standard errors) per dex of metallicity at fixed age, so the two measurements agree in level and in how the heights depend on metallicity. They disagree in how the heights grow. At fixed metallicity our exponential fits over theirs change by $+0.206 \pm 0.029$ (+7.1 standard errors) per ln age, from 0.88 over 6 cells below 2.51 Gyr to 1.19 over 16 cells above 5.01 Gyr (Figure 8). Against the σ_z of the giants of each cell, our exponential heights grow as $\sigma_z^{1.26\pm0.09}$ at fixed metallicity, whereas their published heights grow as $\sigma_z^{0.92\pm0.08}$, consistent with $h_Z \propto \sigma_z$, the growth of an isothermal population in a harmonic potential. The exponential fit to the tracer, however, grows with exponents of 1.47 to 1.52 between 10 and 25 km s^{-1} across the admissible potentials of Section 5.3, and of 1.30 to 1.34 over the whole profile, because the hotter populations reach heights where the potential is no longer harmonic. A potential that is denser or lighter near the plane within the measured errors moves the heights of all the populations by similar factors, which changes their level more than their growth, so the uncertainty of the potential does not account for the slower growth of their heights. The selection of APOGEE does not account for it either: in the cells of LAMOST, with another survey's target-

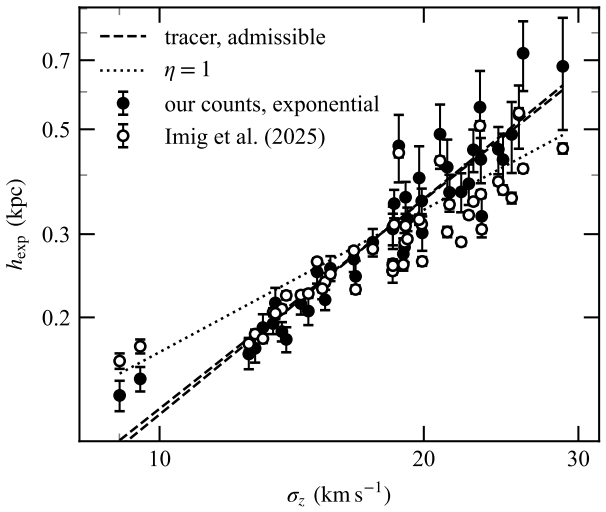


Figure 8. Our exponential fits to the counts of each cell of J. Imig et al. (2025) (filled) and their published heights at the Sun (hollow), against the σ_z of the giants of the cell. The dashed lines give the growth of the exponential fit to the tracer across the admissible potentials, and the dotted line $h_Z \propto \sigma_z$, through the mean of our fits.

ing and ages, our exponential fits over theirs change by $+0.183 \pm 0.019$ (+9.5 standard errors) per ln age.

Part of the difference can come from the range of the fits. Their fit takes the stars at every height that APOGEE reaches, whereas ours stops at 1.2 kpc. For the tracer, the exponential fit to the whole profile is 0.87 to 1.00 times the fit below 1.2 kpc, the lowest for the hottest population, so their old heights should come out thinner than ours by up to that factor. That would account for part of the excess of our old cells, but not for the young cells, where our fits are the thinner. The rest is untested. Both measurements use the same stars on the same age scale, so the errors of the ages mix the same neighbors into both, whereas the two measurements differ in where the stars lie: their height at the Sun comes from a model of the whole footprint, in which each population has one exponential in $|z|$ at every radius and a linear flare, whereas ours comes from the stars within 1.5 kpc of the Sun.

7. DISCUSSION

7.1. *Age-thickness relations*

The heights of mono-age populations grow with age as age$^{\eta\beta_z}$, not as the age$^{2\beta_z}$ of the sheet. Over 0-10 Gyr in the Auriga halos, σ_z grows with exponents β_z of 0.48 to 0.94 and h_Z with exponents β_h of 0.55 to 1.11. The products $\eta\beta_z$ are 0.56 to 1.05, whereas $2\beta_z$ would be 0.96 to 1.88. In the Milky Way our counts give β_z of 0.42 to 0.48 and β_h of 0.46 to 0.53, against $\eta\beta_z$ of 0.45 to 0.53 and $2\beta_z$ of 0.83 to 0.96. Because η and β_h come from the same heights, their agreement is a consistency of the fits rather than a test. The test is that η follows the local slope of the tracer of each potential (Section 3.3). The sheet overstates the growth of the heights by a factor of $2/\eta$, about 1.6-1.8 here. M. Martig et al. (2026, Sect. 5.3) expected from the sheet that the half-mass heights of the stars of lenticular galaxies would grow with age with an exponent between 0.8 and 1, and found otherwise. A tracer lowers that expectation by the factor $2/\eta$, although no value of η produces the flat half-mass heights of their younger populations, so the tracer does not by itself account for their result.

7.2. *The shape of coeval populations*

Two departures from isothermality appear in this work, in opposite directions. In Auriga, the dispersion of a population grows with height, even within 0.25 Gyr of ages, which puts more mass far from the plane than the isothermal profile of its midplane dispersion (Section 3.5) and gives its vertical actions heavier tails than an isothermal population (Section 3.7). In the Milky Way, Y.-S. Ting & H.-W. Rix (2026) find actions with a sharper cutoff, $b = 0.51$, which they attribute to heating by molecular clouds. S. Sharma et al. (2021) find that the dispersions of stars of fixed age and metallicity rise with height, whereas J. Bovy et al. (2012a) find mono-abundance populations nearly isothermal in $|z|$. Our star counts do not separate these shapes: the population with $b = 1/2$ fits them in a potential denser at the midplane than measured, close to that of P. J. McMillan (2017), whereas the isothermal tracer fits them at the measured density. For the heights, the shape is a correction of order 10 percent at fixed σ_z, small against the factor $2/\eta$ by which the sheet misses, whereas for the dynamics of disk heating the shape is the record that Y.-S. Ting & H.-W. Rix (2026) read.

7.3. *The radial action and the scale height*

For a population of one age, the slope $a = J_{R,\rm med}/h_Z$ of the relation between radial action and scale height grows with age as age$^{2\beta_R - \eta\beta_z}$ for a tracer in a fixed potential, rather than the age$^{2(\beta_R - \beta_z)}$ of the sheet at fixed Σ, so the derivation of the slope through the sheet does not carry over to single populations. Across galaxies, the heights of whole disks follow from the tracers of their populations (Section 4). With a reproducible proxy for the radial action of P. A. Palicio et al. (2025), read from maps of the median J_R as they describe them, the slope against the self-consistent heights of the seven halos is 72 ± 28 km/s, against their 72.6 km s^{-1}. It depends, however, on one halo, Au2, without which it is

25 ± 15 km/s, so seven halos do not test the relation across galaxies.

7.4. *Limitations*

The Auriga snapshots are at the standard resolution of the public release, whose stellar softening is comparable to the heights of the youngest populations. R. Pakmor et al. (2025) find that the rise of the velocity dispersion at large radii seen at that resolution might be entirely numerical. Numerical heating raises the dispersion and the height of a population together, so it shifts both sides of the test, but it could change η and the shape of the actions, which the Level 3 runs of the release, at higher resolution, would test. Every Auriga population comes from one annulus per halo, so we do not know how η depends on radius. The fitted potential is axisymmetric, which the potential of the snapshot averaged over azimuth checks but cannot make exact for the bar of Au23. In the Milky Way, the family of potentials varies the thin disk and the halo of P. J. McMillan (2017) but neither its thick disk nor its gas, and is not refitted to the rotation curve. The surface density of J. Bovy & H.-W. Rix (2013) comes from stars modeled in equilibrium, so the agreement of the tracer with the counts remains one of consistency with that assumption. The inversion constrains ρ_0 but not $\Sigma_{1.1}$. The Jeans equation of the Milky Way leaves out the tilt term, which held up the old populations of Au19. Finally, the slower growth of the heights of J. Imig et al. (2025) remains unexplained.

8. CONCLUSIONS

1. In seven Auriga disks, the isothermal tracer in a potential fitted to all the particles predicts the scale heights of 63 of 70 populations of 1 Gyr within 10 percent, with a scatter of 6.2 percent, just above the floor of 5.4 percent that the vertical Jeans equation with the measured dispersion profile and tilt term sets. The self-gravitating sheet does so for 8, making young populations too thin and old ones too thick.

2. The heights grow as σ_z^η with η of 1.10 to 1.26, which equals within two standard errors the local slope of the tracer of the potential in 6 of 7 halos, so age-thickness relations grow as $\text{age}^{\eta\beta_z}$ rather than $\text{age}^{2\beta_z}$.

3. The populations are not exactly isothermal: a residual over 2-10 Gyr that persists with the disk stars alone and in bins of 0.25 Gyr, an excess of mass far from the plane, and vertical actions with heavier tails than an isothermal population all come from dispersions that rise with height, which the Jeans equation captures.

4. Summed over ages, the tracers predict 0.92 to 1.04 of the heights of whole disks and follow their flare.

5. In the solar neighborhood, Gaia DR3 star counts split by LAMOST and APOGEE ages follow the tracer with no trend with age, at medians of 0.97 to 1.07 in potentials that match the measured local densities, and imply midplane densities of 0.089 ± 0.017, 0.104 ± 0.013 and $0.120 \pm 0.030\, M_\odot\, \text{pc}^{-3}$. Doubling the errors of the ages leaves these results in place. Splitting the age tracers at their median magnitude moves no LAMOST height by more than two standard errors.

6. The published heights of J. Imig et al. (2025) agree with our counts in level and in their dependence on metallicity once both are fitted the same way, but grow with σ_z more slowly than the tracer in any admissible potential, for reasons that our data cannot reach.

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Guoshoujing Telescope (the Large Sky Area Multi-Object Fiber Spectroscopic Telescope LAMOST) is a National Major Scientific Project built by the Chinese Academy of Sciences. Funding for the project has been provided by the National Development and Reform Commission. LAMOST is operated and managed by the National Astronomical Observatories, Chinese Academy of Sciences. We use the LAMOST-Gaia red-clump catalog of Y. Huang et al. (2020).

We use the density fits of J. Imig et al. (2025), released under CC BY 4.0 at https://github.com/astrojimig/mw_density_imig2025.

APPENDIX

A. THE WINDOW OF THE FIT

The scale height of a profile that is not exponential depends on the window in which it is fitted. A $\mathrm{sech}^2(z/z_0)$ layer is flat for $|z| \ll z_0$ and falls as $\exp(-2|z|/z_0)$ above, so a window much wider than z_0 returns a scale close to the mean height of the whole profile, $z_0 \ln 2$, whereas a window narrower than z_0 holds an almost flat profile, which the fit reads as a very large scale, or as no scale at all when the mean height reaches $L/2$. We drew 40 samples of 5,000 heights from an exponential, a sech^2 layer, and a mixture of two exponentials, and fitted each in a fixed window with $Z_{\rm lim} = 1\,$kpc and in the self-consistent window $L = 2h_Z$ (Figure 9). The fixed window recovers exponentials within 2.6 percent of their scale, as the maximum-likelihood estimate of an exponential should, whereas it returns 0.73 times the self-consistent height of a sech^2 layer that is thin against the window and 1.66 times that of one as thick as the window. The self-consistent window recovers every injected profile within 2.3 percent. The Auriga populations follow the sech^2 curve: below $h_Z = 0.5\,Z_{\rm lim}$ their median ratio of the fixed to the self-consistent height is 0.91, against 0.89 for sech^2 and 1.00 for an exponential, as expected of tracers in a potential that is harmonic near the plane. The windows of P. A. Palicio et al. (2025, Table 1), with $Z_{\rm lim}$ of 0.5 to 2.5 kpc, are fixed in this sense.

The self-consistent window takes the smallest L that satisfies $L = 2h(L)$. With few stars inside $L/2$, the profile of a hot population is nearly flat inside a small window, so that the noise of the fitted $1/h$ can satisfy the condition by chance, far below the true scale. In 3000 draws of 300 stars from an exponential of $h = 3\,$kpc, a minimum of 20 stars inside $L/2$ gave such a root in 25 draws, a minimum of 50 in 5 and a minimum of 100 in none, which is the minimum we use.

B. THE MOCK GALAXY AND THE POTENTIAL

We tested the analysis on a mock snapshot of a known axisymmetric galaxy, written in the units and the file layout of Auriga, tilted, moving, and close to two faces of the periodic box. The analysis recovers the spin axis within 0.011 degrees, the circular velocity of the annulus within 0.3 percent, $\Delta\Phi$ at the median radius within 1.0 percent over $0.05 < z \leq 1.5\,$kpc, and the height and the median radial action of every bin within 0.2 percent and 0.6 percent of the truth. In the real snapshots, the potential averaged over eight sectors of azimuth differs from the axisymmetric fit by

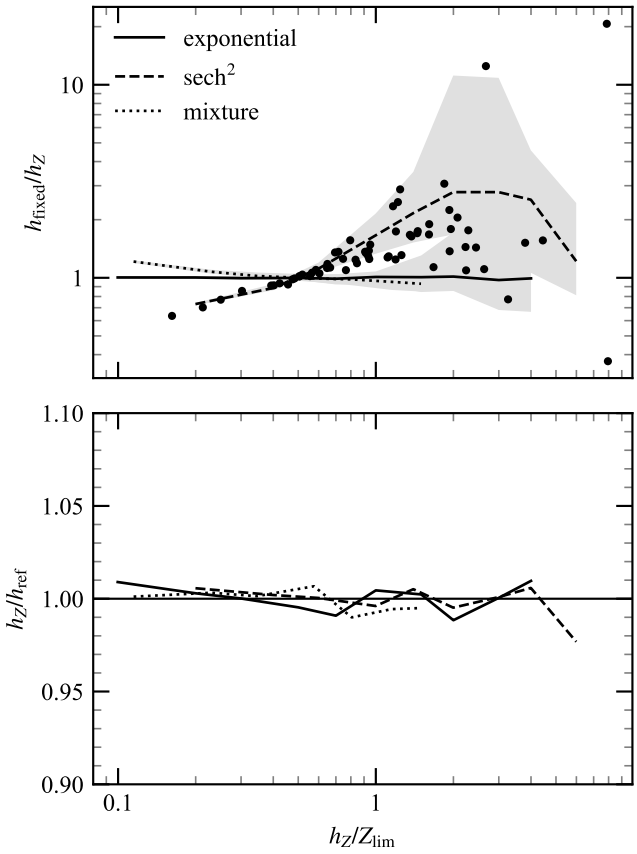


Figure 9. Top: the height fitted in a fixed window $Z_{\rm lim} = 1\,$kpc over the self-consistent height, against $h_Z/Z_{\rm lim}$, for injected exponentials (solid), sech^2 layers (dashed) and mixtures of two exponentials (dotted), with gray bands between the 16th and 84th percentiles of the draws, and for the Auriga populations (dots). Bottom: the self-consistent height of the injected draws over the height of the exact profile.

-68 to $44\,(\mathrm{km\,s^{-1}})^2$ between the plane and $z = 1\,$kpc, so that the tracer in the potential of the snapshot is 0.97 to 1.04 times the tracer in the fit, even with the bar of Au23.

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